Challenges and Solutions Around the Megawatt Charging System

Developing and Testing of Systems With Charging Capacities up to 3.75 MW

Charging at very high power levels is an inevitable requirement for the electrification of heavy traffic, ships, aircraft and construction and transport machinery, for example in mining. The MCS (Megawatt Charging System) will serve as the spanning standard for charging these target groups. At the moment, vehicle and infrastructure manufacturers require not only expertise in the underlying technologies but also flexible development and testing tools that can meet the demands of MCS.

Unlike in many electric passenger transport, the focus for large electric vehicles and machinery is primarily on economic efficiency. These are almost exclusively commercially operated machines, for which maximum availability is a key requirement. This means that the charging time needs to be minimized, which leads to an enhancement of the charging capacity. A look at the battery capacities of large vehicles reveals that, unlike DC charging according to the European CCS standard (Combined Charging System), the required charging power should ideally be higher by a factor of about 10 to 15 (see figure 1). The MCS standard that is currently in standardization is aiming, for example, bringing the batteries of heavy traffic to a charge status of 80 % in approximately 45 minutes. This is maintainable by the specified 3.75 MW. Additionally, this time span harmonizes well with the driving breaks required by law.

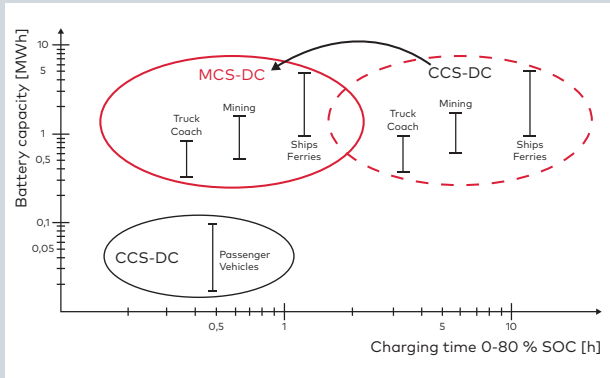

Figure 1: Comparison of battery size and charging times

Charging With up to 3000 A at 1250 V

The charging power of 3.75 MW results from key data at the maximum of 1250 V and 3000 A. In comparison to the standards of GB/T, CHAdeMO, CCS and NACS, MCS real-

izes by the factor of 15 up to 4.2 shorter charging times. Technologically MCS derives from CCS, but they differ drastically from each other in some aspects (see figure 2).

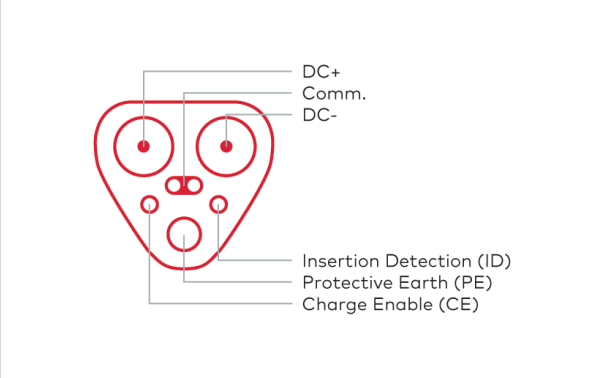

Figure 2: Pin assignment of MCS

The system initially needs a resilient plug-in system, which is capable of transferring electricity up to 3000 A and shield and handle voltages up to 1250 V. It is defined in IEC TS 63379 and describes simultaneous cooling options, which are actively prescribed for example for maximum power. In the low-level communication of CCS, the control pilot (CP) maps the various states of the charging process through PWM-generated voltage levels in the range of ±12 V. With MCS, however the control pilot pin is called Charge Enable (CE) and uses 5 VDC without PWM. The states are represented by lower voltages whose valid level bands are significantly narrower, as are the distances between them. Furthermore, MCS designates the Proximity Pin (PP) as Insertion detection (ID). Like the PP for CCS, the ID pin works with 5 VDC but can also be supplied with 12 V or 24 V to supply auxiliary consumers if necessary. MCS low-level communication is described in IEC 61851-23-3, although the standardization process is still afoot and the standard is expected to be published in 2026 (see figure 3).

High-Level Communication: 10BASE-T1S With Multipoint Connections

The high-level communication is based on ISO 15118-20, whereas Amendement-1, among other things, introduces additonal service IDs for MCS (ISO 15118-20 Amd-1). The communication stack seems similar to CCS and uses TLS, TCP/IP, Ethernet but 10BASE-T1S according to ISO 15118-10 as a physical layer. 10BASE-T1S is a single twisted pair Ethernet with a bandwidth of 10 Mbps half-duplex and Differential Manchester Encoding (DME) based on the IEEE 802.3cg specification. The high-level communication with MCS does not require a crosstalk PLC (Powerline Communication) and the SLAC (Signal Level Attentuation Characterization) procedure used to detect the connected charging situation or counterpart has become obselete. Different to the typical switching ethernet with point to point connection and the resolution of collisions in the switch, 10BASE-T1S executes a system with multipoint connection and Physical Layer Collision Avoidance (PLCA), which was originally developed for automotive use. The infrastructure side operates as a coordinator and is assigned node ID 0, while the electric vehicle is assigned ID 1 (Drop Node 1). Optionally, the communciation bus according to ISO 15118-10 allows for up to six additional nodes (Drop Nodes) according to ISO 15118-10, for example for the connector and inlet socket, to transmit sensor data, current temperatures or other information.

PLCA: Cyclical Data Exchange Without Collisions

Each bus participant is assigned a unique node ID, which it requires for accessing the bus. PLCA opeartes cyclically, allowing each node to the bus once per cycle in ascending order of node IDs. Each node receives a transmit opportunity (TO) within its time slot. This means that a node may only send its frame when it is its slot's turn. At all other times it remains idle and listens to the bus. The latter is necessary in order to maintain the synchronization with the system, as the cycle times may vary in length. The EVSE

	MCS	GB/T	CHAdeMO	CCS Type 2	NACS
Max. Voltage	1250 V	1000	1000 V	1000 V	1000 V
Max. Current*	3000 A	250 A	400 A	500 A	900 A
Max. Charging Power	3750 kW	250 kW	400 kW	500 kW	900 kW
High-Level Communication	ISO 15118-20 Amd-1	GB/T 20234	CHAdeMO	DIN 70121, ISO 15118-2/-20	DIN 70121, ISO 15118-2/-20
PHY Layer for HLC	10BASE-T1S	CAN	CAN	PLC	PLC
Inlet/Outlet Surface	IEC TS 63379	GB/T 20234	IEC 62196-3	IEC 62196-3	SAE J3400

*continuously

Figure 3: Overview of the different charging systems

(Electric Vehicle Suplly Equipment) serves as a coordinator and introduces a cycle by sending a beacon signal. Immediately the slots follow and every node is allowed to send one time. When a specific slot is scheduled, it must begin transmitting its frame within an defined time period (TO_TIME – Transmit Opportunity Time). If the time elapses without an action taken, the node with the next higher ID is allowed to start transmitting. With this procedure, PLCA reduces idle times and ensures optimal use of the available bandwidth, while also decreasing average cycle times.

Test and Development Tools for MCS

Although the MCS standardization is not completed yet, the development of charging systems on OEM and infrastructure has already begun. In combination with the right analysis, simulation and test system, MCS development can start now and be managed effectively. This requires flexibility and timely adaptation of the system to the ongoing standardization process. Early-stage simulations and tests help prevent design errors, as correction processes become increasingly costly the later errors are detected. Ideally, the test system consists of perfectly coordinated software and hardware components. Scalability ensures that costs are kept under control and allows assembling exactly the system needed for specific requirements, desired testing depth, charging power level, and more. One of the key challenges with MCS is that the final test specifications are not yet available. Therefore, tests that have already been created should be easily edited and reusable after changes. Furthermore, the used HIL system must provide the required precision with its measuring technology to reliably identify the narrower voltage bands of MCS low-level communication. Systems previously designed for CCS development may not meet these requirements. Finally, there must be the capability to perform power tests up to 3.75 MW.

Ready for MCS: From Simulation to Performance Tests

Vector's MCS solutions comprise a carefully coordinated system of software and hardware (see figure 4). The central software for analysis, simulation and testing is CANoe option SmartCharging. This system allows full simulation of either the vehicle or the charging station. It provides detailed, time-synchronized analyses of communication and events and can also integrate other bus systems like CAN and LIN communications into the test scenario. The simulation behavior and message content can be customized in many ways, for example regarding the MCS service IDs. Voltages and resistances of the Charge Enable and Insertion Detection pins in low-level communication are also configurable. Injecting faulty messages reveals system responses to undefined or unauthorized communication data. The TLS encryption, mandatory since ISO 15118-20,

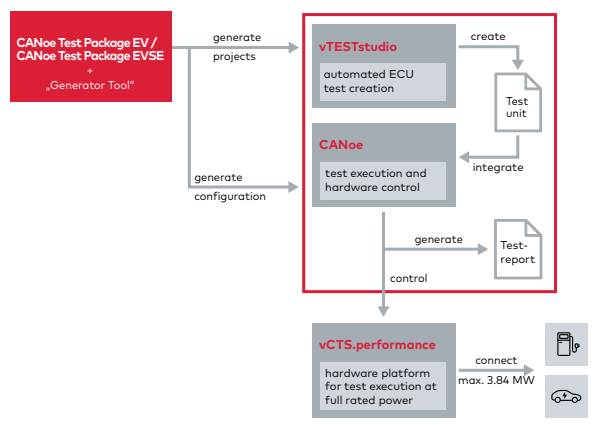

Figure 4: Workflow for testing charging communication between charging stations and electric vehicles

can be optionally activated or deactivated during tests. Vehicle and charging infrastructure manufacturers need specific conformity and interoperability test cases to validate their charging electronics. The two CANoe Test Packages EV and EVSE provide such test cases in a comprehensive test library. With the source code of the Test Packages, testers can also quickly and easily implement custom test sequences that go beyond the standardized test cases. CANoe is responsible for executing the tests, while also handling the hardware control required for test automation and generating the final test report. Until the official MCS test specifications are published, it is recommended to perform conformity tests based on ISO 15118-20. Tests for SLAC, which are not available in MCS, can be easily deactivated.

Scalable DC Power Supply up to 3.84 MW

To execute realistic performance tests in the megawatt range, Vector has developed a highly scalable DC power supply in collaboration with EA Elektro-Automatik. The

Figure 5: With vCTS.performance, performance tests in the megawatt range can be carried out and charging communication can be checked

vCTS.performance delivers up to 1.500 V charging voltage and a maximum charging power of 3.84 MW (see figure 5). A master rack with a base power of 180 kW can be expanded with additional racks, each providing up to 300 kW. This means that those who only want to test with 1 MW or 2 MW do not have to procure the full 3.84 MW system.

Conclusion

In order to jump onto the future MCS market, manufacturers of large electric heavy-duty vehicles, ships, aircraft and construction machinery should start their development efforts as soon as possible. The same applies to companies on the infrastructure side. With the help of suitable development, simulation and testing tools, the underlying complexity, especially of smart charging communication, can be easily managed. The system available from Vector offers high flexibility and scalability, with software and hardware perfectly coordinated. It covers all aspects from low-level to high-level communication and even performance testing at the maximum specified MCS charging power. Vector is also represented in the relevant standardization committees and quickly adapts its solutions to the ongoing MCS standardization process.

Dr. Raphael Pfeil is Product Manager for Development and Test Tools for Smart Charging at Vector Informatik GmbH. He has extensive knowledge of ECU development and currently specializes in conformity and interoperability testing for various smart charging standards. He also contributes his extensive expertise in the field of communication and charging to several standardization committees.

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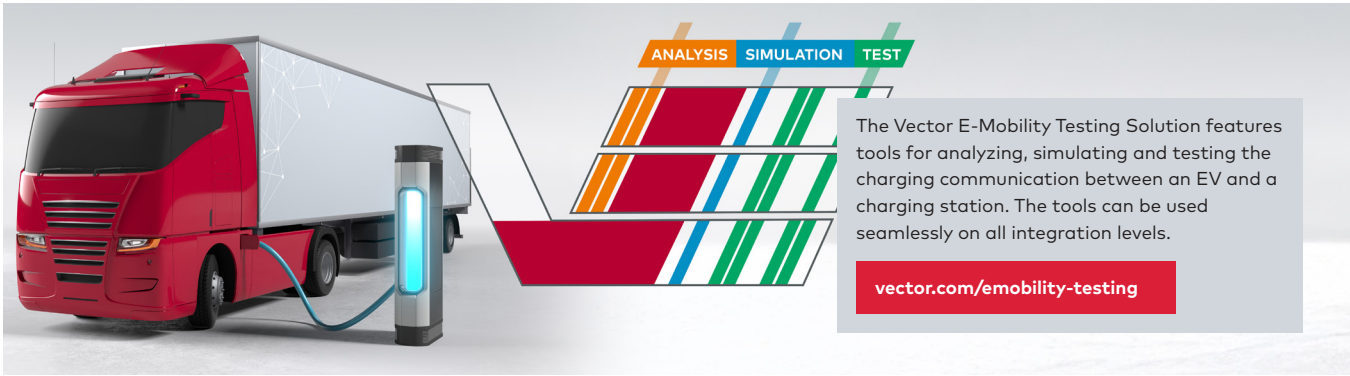
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